

with municipal dashboards to view changes in real time.

VSCP can be integrated with multiple programming platforms (Python, NodeJS, and many others) as well as Raspberry Pi, Arduino, and Home Assistant-based kits for many IoT projects. Detailed documentation on this is available at <https://docs.vscp.org>.

Interfacing VSCP with Node-RED

Node-RED is a flow-based programming tool for wiring together hardware devices, APIs, and online services. It runs on the Node.js platform. To build automation workflows for Industrial IoT systems or smart cities, VSCP events can be processed through the drag-and-drop environment of Node-RED.

NodeJS is a scalable and high-performance JavaScript runtime built for web solutions, and has features for VSCP integration in IoT projects. By leveraging NodeJS programming with Node-VSCP, a very high-performance server-side application can be created that communicates with VSCP networks. NodeJS is ideal for real-time applications because of its async, event-driven model.

To integrate VSCP with Node-RED, the package *node-red-contrib-vscp* is installed so that GUI based scenarios with multiple links can be created. The Node-RED package is available at <https://www.npmjs.com/package/node-red-contrib-vscp>.

Node-VSCP is complemented by an extensive ecosystem as there are assorted packages for HTTP servers, WebSocket communication, and to interact with databases. Developers can work on end-to-end IoT solutions that integrate them with dashboards, APIs, and other externally accessible services.

Custom development or community nodes help communicate with

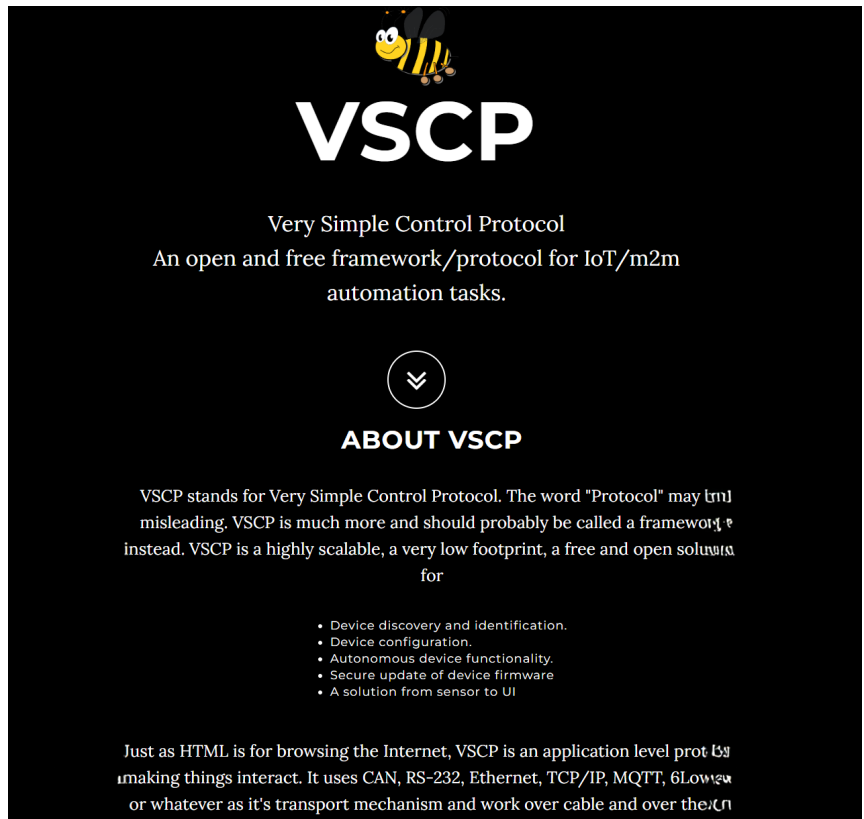


Fig. 2: VSCP

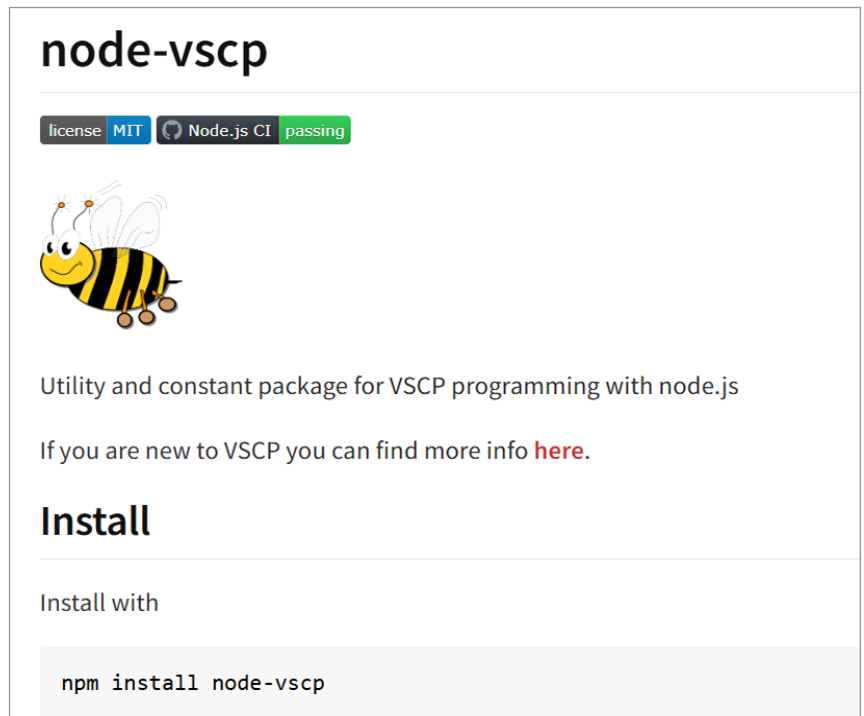


Fig. 3: NodeJS package for VSCP